Machine Learning
Assignment 4

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February 28, 2026

I. Written Assignment

- **Unanswered Questions：**

  - **Week-4 Questions**:

1. Why logistic regression does not use MSE loss?

2. What will happen if we use MSE loss on logistic regression model?

Ans.

In [1], in the case which shows the MSE for estimating $f(0)$:

$$\begin{aligned} MSE(x_0) &= E_\tau[f(x_0) - \tilde{y}_0]^2 \\ &= E_\tau[\tilde{y}_0 - E_\tau(\tilde{y}_0)]^2 + [E_\tau(\tilde{y}_0) - f(x_0)]^2 \\ &= Var_\tau(\tilde{y}_0) + Bias^2(\tilde{y}_0) \end{aligned}$$

Thus as p increases, the estimate tend to be 0 more often than not, and hence the MSE levels of f at 1.0, as does the bias, and the variance starts dropping.

II. Program Assignment

II. A. Classification Model

- **Data Management**
  1. Use package "xml.etree.ElementTree" to parse xml file, obtaining the value of temperature in `<content>`.
  2. Generating grid (representing longitude and latitude), and generating the dataset.
  3. Setting label to 1 if the temperature is valid, otherwise set label to 0.
  4. Combining grid with label, then we have the data with the format (longitude, latitude, label), and save the data with CSV file.
- **Partition Dataset**
  - Training set : 70%
  - Validation set : 15%
  - Testing set : 15%
- **Model Schematic**
  - 2 neurons in the input layer
  - 2 hidden layers, and the 1st hidden layer with 16 neurons while the second with 32 neurons
  - 1 neuron in the output layer
  - Use ReLU activation function between the hidden layer and Sigmoid function follows the output layer.
  - Use BCE loss and Adam's optimizer
- **Result**

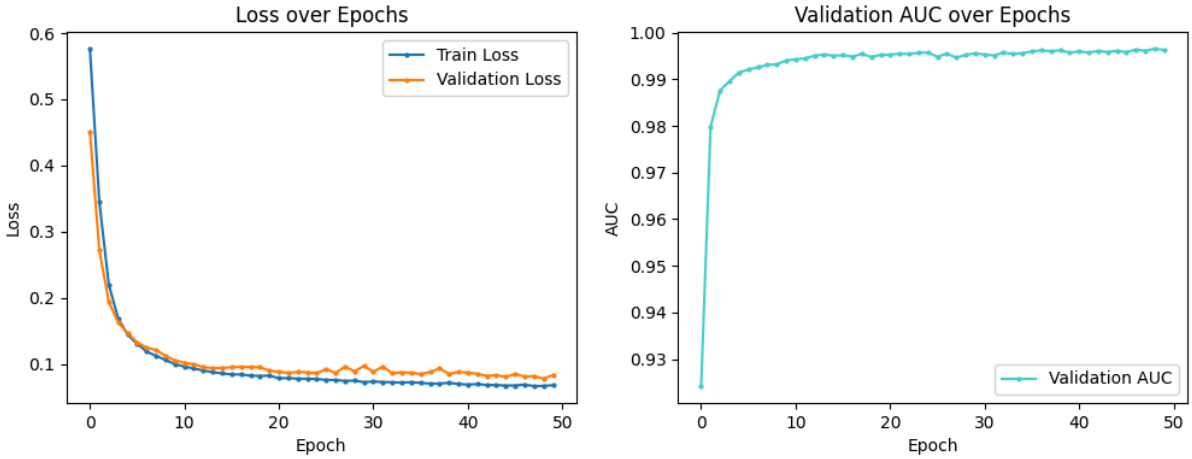

Figure 1. Left is the Loss curve over epochs and right is the graph of validation AUC over epochs.

```
Epoch 26/50 - Train Loss: 0.0741 |  Val Loss: 0.0852 |  Val AUC: 0.9956
Epoch 27/50 - Train Loss: 0.0736 |  Val Loss: 0.0801 |  Val AUC: 0.9962
Epoch 28/50 - Train Loss: 0.0730 |  Val Loss: 0.0801 |  Val AUC: 0.9963
Epoch 29/50 - Train Loss: 0.0729 |  Val Loss: 0.0891 |  Val AUC: 0.9951
Epoch 30/50 - Train Loss: 0.0721 |  Val Loss: 0.0840 |  Val AUC: 0.9963
Epoch 31/50 - Train Loss: 0.0718 |  Val Loss: 0.0792 |  Val AUC: 0.9963
Epoch 32/50 - Train Loss: 0.0703 |  Val Loss: 0.0802 |  Val AUC: 0.9962
Epoch 33/50 - Train Loss: 0.0705 |  Val Loss: 0.0775 |  Val AUC: 0.9965
Epoch 34/50 - Train Loss: 0.0697 |  Val Loss: 0.0838 |  Val AUC: 0.9961
Epoch 35/50 - Train Loss: 0.0692 |  Val Loss: 0.0764 |  Val AUC: 0.9966
Epoch 36/50 - Train Loss: 0.0684 |  Val Loss: 0.0752 |  Val AUC: 0.9967
Epoch 37/50 - Train Loss: 0.0684 |  Val Loss: 0.0794 |  Val AUC: 0.9962
Epoch 38/50 - Train Loss: 0.0684 |  Val Loss: 0.0814 |  Val AUC: 0.9960
Epoch 39/50 - Train Loss: 0.0674 |  Val Loss: 0.0767 |  Val AUC: 0.9965
Epoch 40/50 - Train Loss: 0.0672 |  Val Loss: 0.0823 |  Val AUC: 0.9959
Epoch 41/50 - Train Loss: 0.0665 |  Val Loss: 0.0821 |  Val AUC: 0.9968
Epoch 42/50 - Train Loss: 0.0671 |  Val Loss: 0.0806 |  Val AUC: 0.9965
Epoch 43/50 - Train Loss: 0.0655 |  Val Loss: 0.0798 |  Val AUC: 0.9965
Epoch 44/50 - Train Loss: 0.0665 |  Val Loss: 0.0793 |  Val AUC: 0.9968
Epoch 45/50 - Train Loss: 0.0648 |  Val Loss: 0.0763 |  Val AUC: 0.9967
Epoch 46/50 - Train Loss: 0.0659 |  Val Loss: 0.0768 |  Val AUC: 0.9968
Epoch 47/50 - Train Loss: 0.0643 |  Val Loss: 0.0714 |  Val AUC: 0.9970
Epoch 48/50 - Train Loss: 0.0656 |  Val Loss: 0.0719 |  Val AUC: 0.9968
Epoch 49/50 - Train Loss: 0.0633 |  Val Loss: 0.0784 |  Val AUC: 0.9965
Epoch 50/50 - Train Loss: 0.0643 |  Val Loss: 0.0744 |  Val AUC: 0.9967
---- Test Set AUC: 0.9978 ----
```

Figure 2. The table of training loss and validation AUC over epochs, and test AUC is manifest at the bottom.

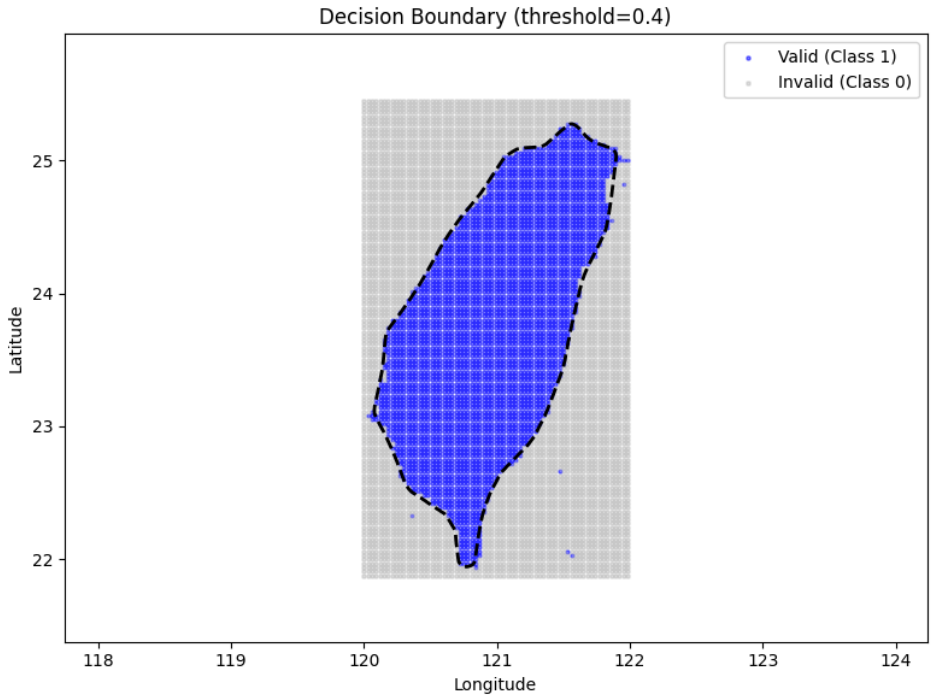

Figure 3. The predicted decision boundary is depicted with black-dash in the figure.

- **Summary** We can see that the model converges well from **Figure 1.** and **Figure 2.**. And the predicted decision boundary in **Figure 3.** also looks great since it quite suitably distinguishes valid points from invalid points.

II. B. Regression Model

- **Data Management**
  1. Use package "xml.etree.ElementTree" to parse xml file, obtaining the value of temperature in `<content>`.
  2. Generating grid (representing longitude and latitude), eliminating the invalid data and set **value** to be the key of temperature.
  3. Combining grid with label, then we have the data with the format (longitude, latitude, value), and save the data with CSV file.
- **Partition Dataset**
  - Training set : 70%

- Validation set : 15%
- Testing set : 15%

- **Model Schematic**
  - 2 neurons in the input layer
  - 2 hidden layers, and there are 16 neurons in each hidden layer
  - 1 neuron in the output layer
  - Use ReLU activation function
  - MSE loss as loss function and Adam's optimizer

- **Result**

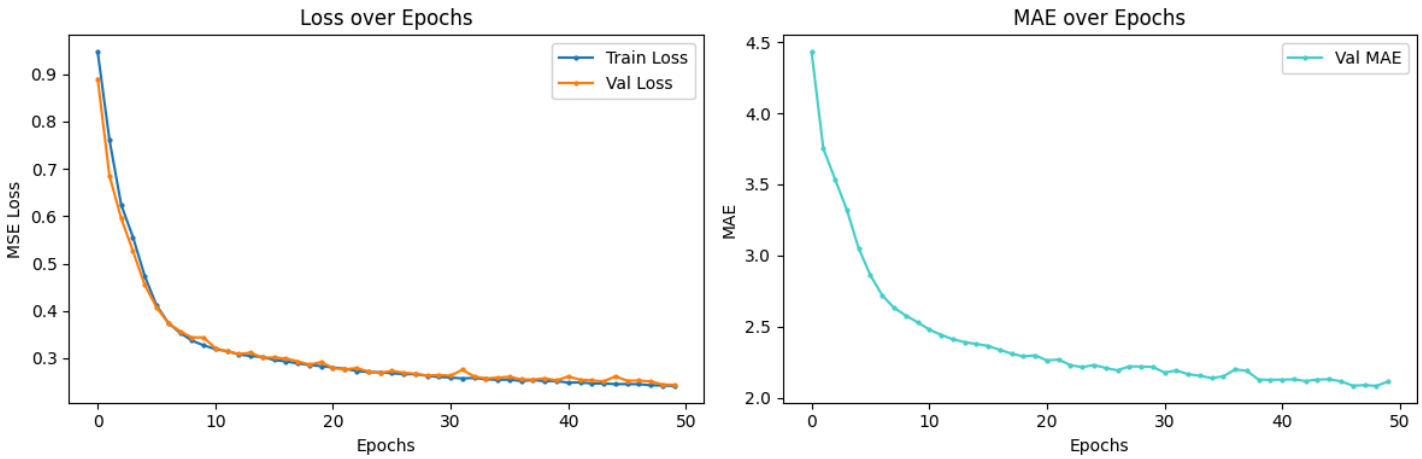

Figure 4. Left is the loss curve over epochs while right is the MAE between predicted temperature and real temperature over epochs.

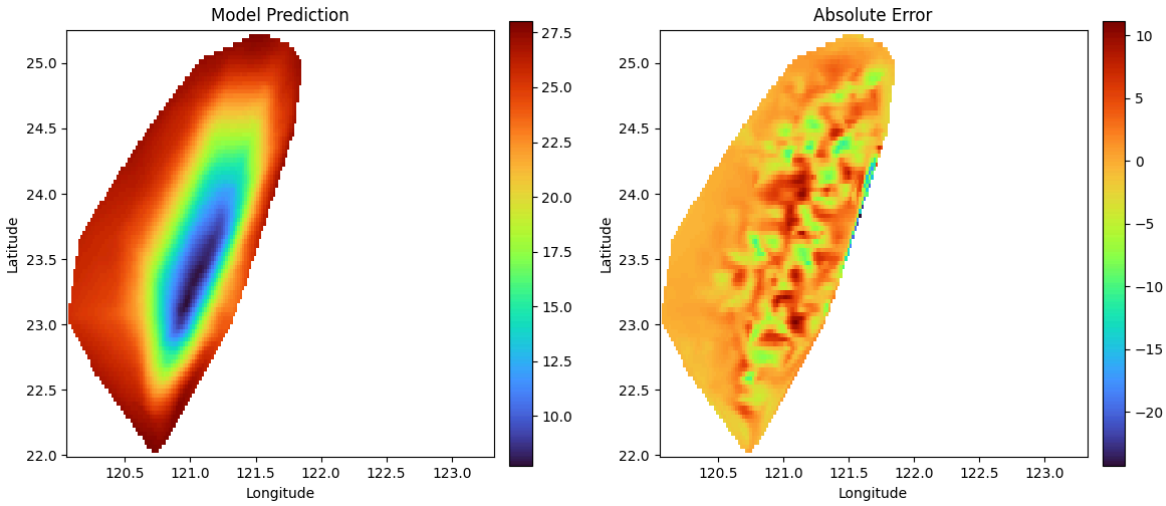

Figure 5. This figure show the predicted temperature on the left and the error between real temperature and predicted temperature on the right.

- **Summary** Though the training loss curve converges, but the MAE is still higher than 2 degrees Celsius. This seems unlike a good prediction nor a good model. I think the main reason why this experimental result occurred is the size of the training dataset is too small to learn. I think the method to improve this is using the dataset with time period last more than a season rather than a moment of a day.

III. Reference

[1] Trevor Hastie, Robert Tishirani, Jerome Friedman. The Elements of Statistical Learning.